Introduction

What is python?

It is high level and general purpose programming language.

What is program?

It is a set of instruction or command to perform specific task.

What is programming language?

It is a medium to communicate with machine

There are 2 types of programming language

- High level language: It is a language which can be understandable by humans. E.X: python, java
- Low level language: it will contain only 0's and 1's it is a binary language which are understandable by machine only not by humans.

Features of python:

- **Easy to read:**

  Learning Python is quite simple. Python's syntax is really straightforward. The code block is defined by the indentations rather than by semicolons or brackets.
- **Easy to code:**

  Python is very easy to learn the as compared to other languages like C, C#, JavaScript, Java, etc. It is very easy to code in the Python language and anybody can learn Python. It is also a developer-friendly language.

- **Free and open source:**

  Python language is freely available at the official website and you can download it. Since it is open-source, this means that source code is also available to the public. So we can download it as well as use it.
- **High level language:**

  It can be understandable by the human.
- **Platform independent:**

  Python language is also a platform independent. For example, if we have Python code for Windows and if we want to run this code on other platforms such as Linux, UNIX, and Mac then we do not need to change it, we can run this code on any platform.

- **Interpreted language:**

  Python is an Interpreted Language because Python code is executed line by line at a time this makes it easier to debug the code.
- **Dynamically typed language:**

  Python is a dynamically-typed language. That means the type for a variable is decided at run time not in advance because of this feature we don't need to specify the type of variable.
- **Numbers of lines of codes are less so it is more efficient.**
- **No prior coding knowledge is required.**
- **Large standard library functions.**

LIBRARY FUNCTIONS:

- Library functions are those whose task is pre-defined by the developer. We can use these library functions but cannot modify its original functionalities.
- It contains bundles of code that can be used repeatedly in different programs.
- It makes Python Programming simpler and convenient for the programmer.
- Python libraries play a very vital role in fields of Machine Learning, Data Science etc.

There are 3 types of library functions

- **Keywords**
- **Inbuilt-functions**
- **Operators**

Keywords:

These are some pre-defined and reserve words in python that have a special meaning.

or

These are the universal standard words whose task is pre-defined by the developers.

- **Universal standard words:** These are the words whose functionality will be the same in all other program.
- To print the keywords on the screen we make use of the syntax

```
import keyword
keyword.kwlist
```

Rules of the keywords:

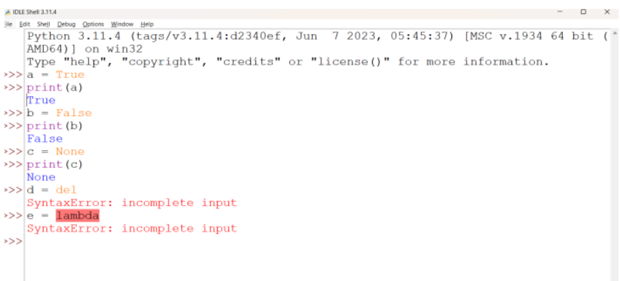
- Keywords cannot be used as variable.
- All the keywords in python should begin with lowercase except ‘True’, ‘False’ and ‘None’.
- ‘True’, ‘False’ and ‘None’ can be used as value to any variable.

Note:

‘True’, ‘False’ and ‘None’ are considered as special keywords because these starts with uppercase character and also only these words can be assigned as value to any variable whereas other cannot be assigned.

E.X: a = True (it is allowed)

b = def (it will throw an error)



```
Python 3.11.4 (tags/v3.11.4:d2340ef, Jun  7 2023, 05:45:37) [MSC v.1934 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> a = True
>>> print(a)
True
>>> b = False
>>> print(b)
False
>>> c = None
>>> print(c)
None
>>> d = del
SyntaxError: incomplete input
>>> e = lambda
SyntaxError: incomplete input
>>>
```

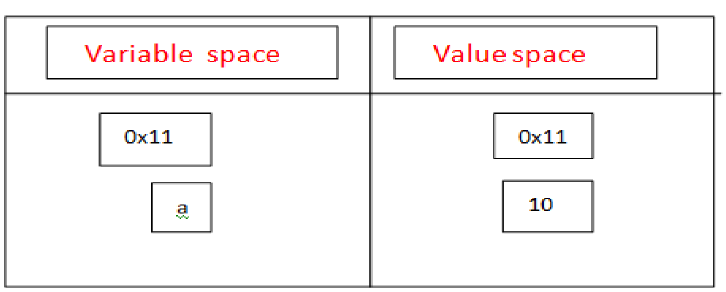
Variable:

It is a name given to the memory location where the value is stored.

Syntax: variable_name = value

Memory allocation for variable in python:

- As soon as control see a variable creation process it divide the memory into two spaces.
  - Variable space
  - Value space
- First, the value will be picked and stored inside a memory block in value space. An address will be given to this which gets stored with respect to variable name in variable space.
- The address given will be in the form of hexadecimal number. But while printing it will get an integer number.
- Example : a = 10



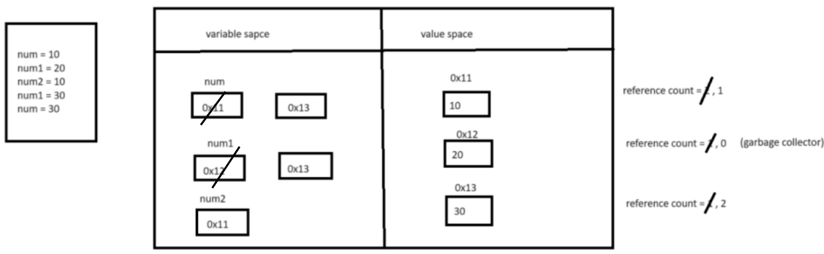
id functions [id(variable/value)] :

It is a function which is used to get the memory address of any variable.

- **id (variable/value)** →it will give memory address in the form of integer.

✓ If we create two variables with the same value then the address of the both the variable will be the same.

✓ When we create same variable with two different values then the address of the new value will stored with respect to the variable.



Reference count:

It is a number of references that point to an object. When reference count becomes zero it become un-reference able and its memory get freed up.

Garbage collector:

It is a memory management technique in python which used to automatically reclaim memory that no longer in use.

- It helps to prevent more consumption of memory.
- Optimize the memory.
- Ensure efficient memory allocation.

Multiple variable creation:

The process of creating multiple variables in a single line is called 'multiple variable creations'.

Syntax:

var1, var2, var3….. = val1, val2, val3…..

Note:

The number of variable should be always equal to the number of values.

```
Python 3.11.4 (tags/v3.11.4:d2340ef, Jun  7 2023, 05:45:37) [MSC v.1934 64
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>
>>
>> num1, num2, num3 = 1, 2, 3
>> print(num1)
   1
>> print(num2)
   2
>> print(num3)
   3
>>
>>
>> num1, num2 = 1, 2, 3
   Traceback (most recent call last):
     File "<pyshell#8>", line 1, in <module>
       num1, num2 = 1, 2, 3
   ValueError: too many values to unpack (expected 2)
>>
>>
>> num1, num2, num3 = 1, 2
   Traceback (most recent call last):
     File "<pyshell#11>", line 1, in <module>
       num1, num2, num3 = 1, 2
   ValueError: not enough values to unpack (expected 3, got 2)
>>
```

Rules of variables:

1. The variables should not be keywords.
2. The variables should not start with numbers.
3. The variables shouldn’t have any special characters except underscore.
4. The variables should not start with space or have space in between.
5. The variables can be alphabets (uppercase or lowercase) or alphanumeric.
6. The length of the variables shouldn’t exceed 32. Even if it exceeds 32 control will not throw any error but as per the industrial standard rules the length should be 32.